Evaluation runs: **nbuyer** (n-buyer-kragl)

generated 2026-11-05 from brain-data a1150b5a2 2026-11-05 14:21:32 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	24	24	conforms	0.0	2	1 h, 250 GB	24	300	brain03, 2026-11-01
3	192	192	conforms	0.3	2	1 h, 250 GB	168	15348	brain03, 2026-11-01
4	1920	1920	conforms	25.5	8	1 h, 250 GB	1344	1368282	brain03, 2026-11-01
5	23040	23040	conforms	1405.8	136	1 h, 250 GB	11819	122224448	brain03, 2026-11-01
6	322560		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

stateless: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	24	24	6	violation	0.0	2	1 h, 250 GB	24	387	brain03, 2026-11-01
3	192	192	96	violation	0.6	2	1 h, 250 GB	192	28380	brain03, 2026-11-01
4	1920	1920	1320	violation	61.5	10	1 h, 250 GB	1920	3361944	brain03, 2026-11-01
5	23040	23040	18720	violation	7083.0	132	48 h, 250 GB	23040	504590220	brain04, 2026-11-01

Not run (stateless). skipped: a smaller size had reached its limit: 6 no-subtract bag, 7 correct bag, 7 no-subtract bag, 8 correct bag, 8 no-subtract bag, 9 no-subtract bag, 9 correct bag, 10 correct bag, 10 no-subtract bag, 11 no-subtract bag, 11 correct bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	24	24	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-01
3	192	192	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
4	1920	1920	conforms	0.2	12	1 h, 250 GB	brain02, 2026-11-01
5	23040	23040	conforms	7.9	148	1 h, 250 GB	brain02, 2026-11-01
6	322560	322560	conforms	2954.0	3227	1 h, 250 GB	brain02, 2026-11-01
7			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-01

stateful: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	24	24	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-01
3	192	192	96	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
4	1920	1920	1320	violation	0.3	16	1 h, 250 GB	brain02, 2026-11-01
5	23040	23040	18720	violation	13.8	410	1 h, 250 GB	brain02, 2026-11-01
6	322560	322560	287280	violation	10154.0	4747	48 h, 250 GB	brain04, 2026-11-02

Not run (stateful). skipped: a smaller size had reached its limit: 7 no-subtract bag, 8 correct bag, 8 no-subtract bag, 9 no-subtract bag, 9 correct bag, 10 correct bag, 10 no-subtract bag, 11 no-subtract bag, 11 correct bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	192	192	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
4	1920	1920	conforms	0.2	10	10 min, 16 GB	0.1	0.1	0.1	in-memory	brain05, 2026-11-02
5	23040	23040	conforms	5.7	86	10 min, 16 GB	0.6	1.0	4.6	in-memory	brain05, 2026-11-02
6	322560	322560	conforms	1886.7	892	1 h, 250 GB	3.9	7.9	1878.8	in-memory	brain05, 2026-11-03
7	5160960	5160960	(1 h)			1 h, 250 GB	27.6	60.4		in-memory	brain05, 2026-11-03
8	92897280	92897280	(1 h)			1 h, 250 GB	421.2	1070.3		in-memory	brain05, 2026-11-03

offline: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	6	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11
3	192	192	96	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11
4	1920	1920	1320	violation	0.3	10	10 min, 16 GB	0.1	0.2	0.1	in-memory	brain05, 2026-11
5	23040	23040	18720	violation	15.1	119	10 min, 16 GB	0.6	1.2	13.8	in-memory	brain05, 2026-11
6	322560	322560		(1 h)			1 h, 250 GB	3.9	8.4		in-memory	brain05, 2026-11
7	5160960	5160960		(1 h)			1 h, 250 GB	27.6	78.7		in-memory	brain05, 2026-11
8	92897280	92897280		oom (250 GB)			1 h, 250 GB	421.2	1459.2		in-memory	brain05, 2026-11

Not run (offline). planned but never scheduled (grid points beyond the ladder): 8 no-subtract bag, 8 correct bag.
skipped: a smaller size had reached its limit: 9 no-subtract bag, 9 correct bag, 10 correct bag, 10 no-subtract bag, 11 no-subtract bag, 11 correct bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	conforms	3.4	160	10 min, 16 GB	0.0	3.4	0.0	in-memory	brain01, 2026-11-02
3	192	192	conforms	4.0	176	10 min, 16 GB	0.0	4.0	0.0	in-memory	brain01, 2026-11-02
4	1920	1920	conforms	8.0	578	10 min, 16 GB	0.1	8.0	0.1	in-memory	brain01, 2026-11-02
5	23040	23040	conforms	51.5	1128	10 min, 16 GB	0.6	46.7	4.8	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	6	violation	3.3	162	10 min, 16 GB	0.0	3.3	0.0	in-memory	brain01, 2026-11-02
3	192	192	96	violation	4.0	155	10 min, 16 GB	0.0	4.0	0.0	in-memory	brain01, 2026-11-02
4	1920	1920	1320	violation	7.8	601	10 min, 16 GB	0.1	7.7	0.1	in-memory	brain01, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
5	23040	23040	18720	violation	58.4	1396	10 min, 16 GB	0.6	46.5	11.9	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	conforms	3.1	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
3	192	192	conforms	3.5	180	10 min, 16 GB	3.5	0.1	0.0	in-memory	brain01, 2026-11-01
4	1920	1920	conforms	5.0	240	10 min, 16 GB	5.0	0.1	0.1	in-memory	brain01, 2026-11-01
5	23040	23040	conforms	23.4	586	10 min, 16 GB	18.9	1.0	4.5	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	6	violation	3.1	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01
3	192	192	96	violation	3.5	180	10 min, 16 GB	3.5	0.1	0.0	in-memory	brain01, 2026-11-01
4	1920	1920	1320	violation	5.1	240	10 min, 16 GB	5.0	0.2	0.1	in-memory	brain01, 2026-11-01
5	23040	23040	18720	violation	30.9	586	10 min, 16 GB	18.9	1.2	12.1	in-memory	brain01, 2026-11-01

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	24	24	conforms	0.8	150	1 h, 250 GB		brain05, 2026-11-02
3	192	192	conforms	1.3	172	1 h, 250 GB		brain05, 2026-11-02
4	1920	1920	conforms	6.6	673	1 h, 250 GB		brain05, 2026-11-02
5	23040	23040	conforms	71.2	1562	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	24	24	6	violation	0.7	145	1 h, 250 GB	53	brain05, 2026-11-02
3	192	192	96	violation	1.2	170	1 h, 250 GB	56	brain05, 2026-11-02
4	1920	1920	1320	violation	6.1	681	1 h, 250 GB	59	brain05, 2026-11-02
5	23040	23040	18720	violation	76.6	1229	1 h, 250 GB	64	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 2 no-subtract bag, 2 correct bag, 3 correct bag, 3 no-subtract bag, 4 no-subtract bag, 4 correct bag, 5 correct bag, 5 no-subtract bag.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	24	24	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
3	192	192	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
4	1920	1920	conforms	0.2	20	1 h, 250 GB		brain04, 2026-11-01
5	23040	23040	conforms	5.2	226	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	24	24	6	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
3	192	192	96	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
4	1920	1920	1320	violation	0.3	18	1 h, 250 GB	25	brain04, 2026-11-01
5	23040	23040	18720	violation	7.9	240	1 h, 250 GB	38	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): 2 no-subtract bag, 2 correct bag, 3 correct bag, 3 no-subtract bag, 4 no-subtract bag, 4 correct bag, 5 correct bag, 5 no-subtract bag.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
6	322560	322560	conforms	561.9	16190	48 h, 250 GB	559.5	7.4	2.3	merge	brain01, 2026-11-04

twins offline, Rust implementation against its Java twin: pair **no-subtract**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
6	322560	322560	conforms	569.4	17561	48 h, 250 GB	562.8	8.0	6.6	merge	brain01, 2026-11-04

twins offline, Rust implementation against its Java twin: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
6	322560	322560	287280	violation	563.9	16190	48 h, 250 GB	559.5	8.0	4.4	merge	brain01, 2026-11-04

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
6	322560	322560	conforms	561.8	16190	48 h, 250 GB	7.4	559.5	2.2	merge	brain01, 2026-11-04

twins offline, Java implementation against its Rust twin: pair **no-subtract**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
6	322560	322560	conforms	569.2	17561	48 h, 250 GB	8.0	562.8	6.4	merge	brain01, 2026-11-04

twins offline, Java implementation against its Rust twin: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
6	322560	322560	287280	violation	567.2	17561	48 h, 250 GB	7.4	562.8	4.4	merge	brain01, 2026-11-04

twins online, Rust implementation (TraceForge) against the Java twin's export

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
6	322560	322560	conforms	1306.6	4857	48 h, 250 GB		brain01, 2026-11-04

twins online, Rust implementation (TraceForge) against the Java twin's export: pair **no-subtract**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
6	322560	322560	conforms	9859.0	14102	48 h, 250 GB		brain01, 2026-11-04

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	conforms	3.4	158	10 min, 16 GB	3.3	0.1	0.1	in-memory	brain02, 2026-11-03
3	192	192	conforms	4.0	154	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain02, 2026-11-03
4	1920	1920	conforms	7.9	569	10 min, 16 GB	7.7	0.3	0.2	in-memory	brain02, 2026-11-03
5	23040	23040	conforms	47.2	7284	10 min, 16 GB	42.4	1.4	4.8	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **no-subtract**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	conforms	3.4	154	10 min, 16 GB	3.3	0.2	0.0	in-memory	brain02, 2026-11-03
3	192	192	conforms	4.2	163	10 min, 16 GB	4.2	0.1	0.0	in-memory	brain02, 2026-11-03
4	1920	1920	conforms	7.9	565	10 min, 16 GB	7.7	0.4	0.2	in-memory	brain02, 2026-11-03
5	23040	23040	conforms	76.1	8305	10 min, 16 GB	42.4	1.2	33.7	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	6	violation	3.3	158	10 min, 16 GB	3.3	0.2	0.0	in-memory	brain02, 2026-11-03

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	192	192	96	violation	4.0	154	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain02, 2026-11-03
4	1920	1920	1320	violation	7.8	569	10 min, 16 GB	7.7	0.4	0.1	in-memory	brain02, 2026-11-03
5	23040	23040	18720	violation	55.4	7284	10 min, 16 GB	42.4	1.2	13.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	conforms	3.3	158	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
3	192	192	conforms	4.0	154	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain02, 2026-11-03
4	1920	1920	conforms	7.8	569	10 min, 16 GB	0.3	7.7	0.1	in-memory	brain02, 2026-11-03
5	23040	23040	conforms	47.5	7284	10 min, 16 GB	1.4	42.4	5.1	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **no-subtract**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	conforms	3.3	154	10 min, 16 GB	0.2	3.3	0.0	in-memory	brain02, 2026-11-03
3	192	192	conforms	4.2	163	10 min, 16 GB	0.1	4.2	0.0	in-memory	brain02, 2026-11-03
4	1920	1920	conforms	7.9	565	10 min, 16 GB	0.4	7.7	0.2	in-memory	brain02, 2026-11-03
5	23040	23040	conforms	74.1	8305	10 min, 16 GB	1.2	42.4	31.7	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	24	24	6	violation	3.3	154	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
3	192	192	96	violation	4.2	163	10 min, 16 GB	0.1	4.2	0.0	in-memory	brain02, 2026-11-03
4	1920	1920	1320	violation	7.8	565	10 min, 16 GB	0.3	7.7	0.1	in-memory	brain02, 2026-11-03
5	23040	23040	18720	violation	55.5	8305	10 min, 16 GB	1.4	42.4	13.1	in-memory	brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	24	24	conforms	3.8	162	1 h, 250 GB		brain02, 2026-11-03
3	192	192	conforms	4.0	171	1 h, 250 GB		brain02, 2026-11-03
4	1920	1920	conforms	9.3	597	1 h, 250 GB		brain02, 2026-11-03
5	23040	23040	conforms	49.2	9196	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-subtract**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	24	24	conforms	3.6	159	1 h, 250 GB		brain02, 2026-11-03
3	192	192	conforms	4.3	170	1 h, 250 GB		brain02, 2026-11-03
4	1920	1920	conforms	10.1	645	1 h, 250 GB		brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
5	23040	23040	conforms	48.0	10395	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	24	24	6	violation	3.5	158	1 h, 250 GB	80	brain02, 2026-11-03
3	192	192	96	violation	4.8	168	1 h, 250 GB	103	brain02, 2026-11-03
4	1920	1920	1320	violation	10.3	2209	1 h, 250 GB	182	brain02, 2026-11-03
5	23040	23040	18720	violation	46.0	9809	1 h, 250 GB	257	brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	24	24	conforms	1.4	2	1 h, 250 GB		brain02, 2026-11-03
3	192	192	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
4	1920	1920	conforms	0.2	16	1 h, 250 GB		brain02, 2026-11-03
5	23040	23040	conforms	5.7	248	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-subtract**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	24	24	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3	192	192	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
4	1920	1920	conforms	0.3	20	1 h, 250 GB		brain02, 2026-11-03
5	23040	23040	conforms	26.9	593	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-subtract**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	24	24	6	violation	0.0	2	1 h, 250 GB	2	brain02, 2026-11-03
3	192	192	96	violation	0.0	4	1 h, 250 GB	2	brain02, 2026-11-03
4	1920	1920	1320	violation	0.3	20	1 h, 250 GB	4	brain02, 2026-11-03
5	23040	23040	18720	violation	10.2	274	1 h, 250 GB	41	brain02, 2026-11-03